ARTHUR GAUTIER

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Data & AI Engineer, seeking opportunities in the field of MLE & Data Science, available in January 2027

WORK EXPERIENCE

Data Scientist – Scor

Singapore & Paris, France / October 2024 – Today

At SCOR, a global leader in reinsurance, I joined the Singapore team to spearhead strategic automation and predictive modeling projects. I contributed significantly to SCOR's AIssistant, a Generative AI (GenAI) initiative designed to revolutionize the underwriting process by augmenting human expertise with advanced language model.

- GenAI & OCR-to-LLM Pipelines: Designed and deployed Generative AI pipelines using specialized Prompt Engineering to automatically extract and summarize complex business information from diverse insurance documents.
- SCOR AI Assistant Contribution: Integrated advanced extraction features into the SCOR AIssistant ecosystem, enabling underwriters to process high volumes of unstructured data with increased speed and accuracy.
- Industrial-grade Development: Developed a Python package for mortality risk prediction using Cox and XGBoost models, applied to activity tracking (IoT) data.

Skills: Python – NLP – Domino – Docker – FastApi – Pyspark – Agile environment – Git/GitLab – GitHub Copilot

Data Scientist – Covéa

Niort, France / January – July 2024

Covéa, a leading insurance company, launched a project to automate its electronic document management (EDM) system. As part of a team of five data scientists, I contributed to this large-scale project aimed at analyzing 72 million images representing 4 million customer files. My role focused on the ADR (Automatic Document Recognition) component, where I studied and developed open-source solutions as an alternative to the current managed service with the goal of reducing costs.

- Design and implementation of an ETL process for creating custom databases and training models.
- Development of an alternative to the current managed Custom Vision service using open-source models (YOLO) and Azure services.
- Benchmarked document recognition models and optimized annotated datasets for training.
- Achieved 97% concordance with the production solution on ~100k images while reducing costs by 64%.

Skills: Python – Cost optimization with a FinOps mindset – Big Data – Microsoft Azure – Databricks – Pyspark – Agile environment

AI Research Assistant - Bangkok University

Bangkok, Thailand / May – July 2023

Design of a demonstrator aimed at optimizing the management of food product sales: creation of a scanner capable of automatically detecting and identifying food items present on a meal tray from a photo taken by the scanner and displaying the corresponding price.

- Developed a YOLOv8-based computer vision system for automatic food recognition and pricing from meal tray images.
- Built and augmented a custom dataset using web scraping, image collection, and Roboflow annotation.
- Deployed the solution on Raspberry Pi and built a full-scale prototype with integrated camera, UI, and LED lighting.

Skills: Python – Computer Vision – OpenCV – 3D Printing – Web Scraping – Raspberry Pi – Google Colab

EDUCATION

Bergen University

Bergen, Norway / August – December 2023

- Machine Learning (Random Forest, SVM, PCA, KNN…), Information Theory, Concurrent programming, Norwegian language.
- Notable Project: Implementation of a decision tree (see project).

SeaTech, School of Engineering

Toulon, France / 2021-2024

- Engineering M.Sc. in Data Science & Processing and Information Systems.
- Notable Projects: Digital Chessboard Transcription via Photography (see project). 'Garden Simulator' Video Game in C++ (see project).

Preparatory Class, Descartes

Tours, France / 2019-2021

- Intensive pre-Engineering preparation course for entrance into elite engineering schools. MPSI-MP (Mathematics, Physics) curriculum.
- Project: Fire Evacuation Simulation and Optimized Exit Placement Research (see project).

INFORMATION

Programming languages: Python, MATLAB, C++, SQL, PySpark

Cloud Platforms/Services: Microsoft Azure, Databricks, Google Colab, Git

Languages: English (Fluent) – French (Native) – Spanish (Elementary) – Chinese Mandarin (Learning)

Hobbies and Interests: Running: 400/800m competition, marathon (Strava) – Cinema and music (SensCritique).